THE GYSIN HOMOMORPHISM FACTORS THROUGH RATIONAL EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a coproduct and proper pushforward reduction to a single rational function on an integral space, followed by a two case identification with Chern class caps

Editorial note. The label lemma-gysin-factors names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The Gysin homomorphism attached to a triple consisting of an invertible sheaf, a nonzero meromorphic section and its effective Cartier divisor of zeros, and the compatibility of Gysin pullback with proper pushforward
- Proper pushforward of cycles preserves rational equivalence, so that a vanishing result on a proper cover descends
- The general Gysin and first Chern class compatibility, and the vanishing of the cap product with the first Chern class of the structure sheaf

Source reference. Original reference 02T8: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 3448–3448. Prerequisite pointer invoked at proof line(s) 3733; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02TA: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 3545–3545. Prerequisite pointer invoked at proof line(s) 3752; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02TI: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 3327–3327. Prerequisite pointer invoked at proof line(s) 3763; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02TM: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 3662–3662. Prerequisite pointer invoked at proof

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line(s) 3766; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02S2: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 2176–2176. Prerequisite pointer invoked at proof line(s) 3754; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0B6Y: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 3496–3496. Prerequisite pointer invoked at proof line(s) 3751; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 3731; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let X/B be good. Let $(\mathcal{L}, s, i : D \to X)$ be as in Definition 02T8. The Gysin homomorphism factors through rational equivalence to give a map $i^* : \mathrm{CH}_{k+1}(X) \to \mathrm{CH}_k(D)$.*

Proof. Let $\alpha \in Z_{k+1}(X)$ and assume that $\alpha \sim_{rat} 0$. This means there exists a locally finite collection of integral closed subspaces $W_j \subset X$ of δ-dimension $k+2$ and $f_j \in R(W_j)^*$ such that $\alpha = \sum i_{j,*}\mathrm{div}_{W_j}(f_j)$. Set $X' = \coprod W_i$ and consider the diagram

$$\begin{array}{ccc} D' & \xrightarrow{i'} & X' \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle p} \\ D & \xrightarrow{i} & X \end{array}$$

of Remark 0B6Y. Since $X' \to X$ is proper we see that $i^*p_* = q_*(i')^*$ by Lemma 02TA. As we know that q_* factors through rational equivalence (Lemma 02S2), it suffices to prove the result for $\alpha' = \sum \mathrm{div}_{W_j}(f_j)$ on X'. Clearly this reduces us to the case where X is integral and $\alpha = \mathrm{div}(f)$ for some $f \in R(X)^*$.

Assume X is integral and $\alpha = \mathrm{div}(f)$ for some $f \in R(X)^*$. If $X = D$, then we see that $i^*\alpha$ is equal to $c_1(\mathcal{L}) \cap \alpha$. This is rationally equivalent to zero by Lemma 02TI. If $D \neq X$, then we see that $i^*\mathrm{div}_X(f)$ is equal to $c_1(\mathcal{O}_D) \cap [D]_{n-1}$ in $\mathrm{CH}_k(D)$ by Lemma 02TM. Of course capping with $c_1(\mathcal{O}_D)$ is the zero map. $\square$